

2nd assignment: Exploring a dataset

Instructions

In this 2nd assignment of the course, you will apply the exploratory data techniques that you have learned during the course on a dataset of your choice. You can either pick a dataset from the collection that we will provide (available from 25th May), or you could choose another dataset that you prefer. If you want to pick your own dataset, first check with your teacher (Carl) that it is suitable for this assignment.

Requirements

The assignment must be done individually and has to be submitted as a paper (~7 pages excluding appendix, 1.5 line spaced). You are reasonably free to structure your hand-in report/paper, but it must contain the following parts/aspects:

- Describe (in text) what the dataset is concerned with. Outline an area of inquiry and/or a tentative substantive question (e.g. How do songs cluster together? Are opinions split by party belonging? Are environmental issues a latent variable?). Feel free to include external literature to build a tentative question.
- Use at least two EDA methods that you have learned in this course. A suitable combination is a dimensionality-reduction method followed by clustering, though other combinations might also be acceptable. Check with your teacher if unsure. Note that SQL queries do not constitute an EDA method.
- Justify your choice of method: why these ones? What are their limitations?
- Interpret your results (including dimensions, clusters etc): feel free to reason freely about the results you obtain
- Include a appendix with your code (that must be commented)

Deadline: Friday 12th June 23.59

Format: Approx 7 pages (excluding appendix), PDF format (use your liu username for the document name, for instance: carno10_project2.pdf)

Submit your paper on Lisam (check Submissions section).

Datasets

Datasets will be provided on 25th May (through Lisam). Do note that you may not use the Spotify dataset (music.sqlite) for this assignment. If you want to work with a different dataset, please contact carl.nordlund@liu.se